

InsightVoice AI

Customer Feedback Intelligence using Machine Learning and Natural
Language Processing

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Google Colab Notebook: InsightVoice AI - Google Colab Notebook

1 Abstract

Customer feedback is one of the most valuable sources for understanding customer satisfaction and improving products. However, manually analyzing thousands of textual reviews is time-consuming and inefficient. This project presents **InsightVoice AI**, a sentiment analysis system that automatically classifies Amazon product reviews into Positive, Negative, and Neutral sentiments using Machine Learning and Natural Language Processing (NLP).

The Amazon Product Reviews dataset containing 568,454 reviews was preprocessed using text cleaning techniques and converted into numerical features using TF-IDF Vectorization. Three machine learning models—Multinomial Naive Bayes, Logistic Regression, and Linear Support Vector Machine (Linear SVM)—were trained and evaluated. Among these, Linear SVM achieved the highest accuracy of **89.39%**, making it the best-performing model for customer sentiment classification.

2 Introduction

The rapid growth of e-commerce platforms has resulted in a massive increase in customer-generated reviews. These reviews provide valuable insights into customer experiences, product quality, and overall satisfaction. Due to the large volume of textual feedback, manual analysis becomes impractical.

Sentiment Analysis is an important application of Natural Language Processing that automatically identifies the emotional tone of textual data. It enables businesses to quickly understand customer opinions and make informed decisions.

The objective of this project is to design and evaluate a machine learning-based sentiment analysis system capable of accurately classifying customer reviews into Positive, Negative, and Neutral categories.

3 Objectives

The primary objectives of this project are:

- Analyze customer reviews using exploratory data analysis.
- Clean and preprocess textual data.
- Convert text into numerical features using TF-IDF.
- Train multiple machine learning classifiers.
- Compare model performance using standard evaluation metrics.
- Select the most accurate model for sentiment prediction.

4 Dataset

The project uses the **Amazon Product Reviews Dataset**, which contains customer reviews collected from various product categories. Each review is classified into one of three sentiment classes based on customer ratings.

Table 1: Dataset Summary

Parameter	Value
Dataset	Amazon Product Reviews
Total Reviews	568,454
Features	10
Sentiment Classes	Positive, Negative, Neutral
Classification Type	Multi-Class

The sentiment distribution is shown below.

Table 2: Sentiment Distribution

Sentiment	Reviews
Positive	443,777
Negative	82,037
Neutral	42,640

5 Methodology

The proposed system follows a standard machine learning pipeline for sentiment classification. The workflow begins with collecting customer reviews from the Amazon Product Reviews dataset. The textual data is then preprocessed to remove unnecessary information and improve data quality. After preprocessing, TF-IDF (Term Frequency-Inverse Document Frequency) Vectorization is applied to convert text into numerical feature vectors. Finally, three machine learning algorithms are trained and evaluated using standard classification metrics.

5.1 Data Preprocessing

The following preprocessing techniques were applied before training the models:

- Conversion of text to lowercase.
- Removal of punctuation and special characters.
- Removal of stop words using the NLTK library.
- Lemmatization of words.
- Removal of extra white spaces.

These preprocessing steps improve the quality of textual data and help the models learn meaningful patterns.

5.2 Feature Extraction

Machine learning algorithms require numerical input; therefore, the cleaned text was transformed using **TF-IDF (Term Frequency-Inverse Document Frequency)** Vectorization. TF-IDF assigns higher importance to informative words while reducing the influence of frequently occurring words, resulting in an effective representation of customer reviews.

5.3 Machine Learning Models

Three supervised learning algorithms were implemented and compared:

- Multinomial Naive Bayes
- Logistic Regression
- Linear Support Vector Machine (Linear SVM)

Their performance was evaluated using Accuracy, Precision, Recall, and F1-Score.

6 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was carried out to understand the characteristics of the dataset before model training. The visualizations below summarize the distribution of ratings and sentiment classes.

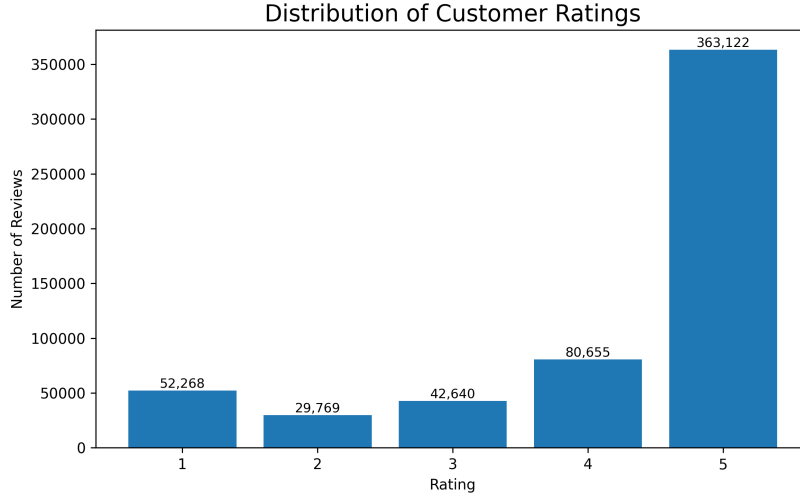


Figure 1: Distribution of Customer Ratings

Figure 1 shows that higher customer ratings occur more frequently, indicating that the dataset contains a larger proportion of positive feedback.

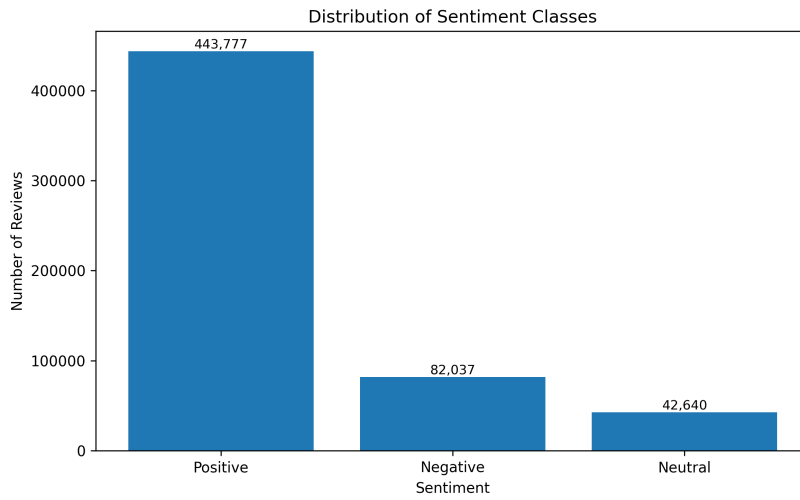


Figure 2: Distribution of Sentiment Classes

Figure 2 illustrates the distribution of Positive, Negative, and Neutral reviews. The dataset is dominated by Positive reviews, which reflects typical customer behavior on e-commerce platforms.

7 Results and Discussion

The performance of the three machine learning models was evaluated using Accuracy, Precision, Recall, and F1-Score. Table 3 summarizes the obtained results.

Table 3: Performance Comparison of Machine Learning Models

Model	Accuracy	Precision	Recall	F1-Score
Multinomial Naive Bayes	84.63%	82.81%	84.63%	80.88%
Logistic Regression	89.29%	88.01%	89.29%	88.20%
Linear SVM	89.39%	88.18%	89.39%	88.17%

Among the evaluated models, the **Linear Support Vector Machine (Linear SVM)** achieved the highest overall accuracy of **89.39%**. Its strong performance across all evaluation metrics demonstrates its effectiveness in classifying customer reviews represented using TF-IDF features.

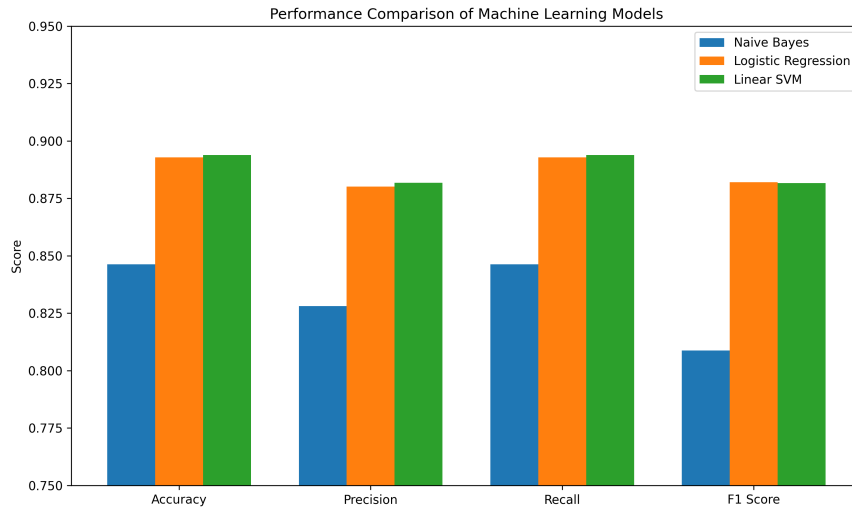


Figure 3: Performance Comparison of Machine Learning Models

The comparison graph clearly shows that Logistic Regression and Linear SVM outperform Multinomial Naive Bayes, with Linear SVM providing the best overall performance.

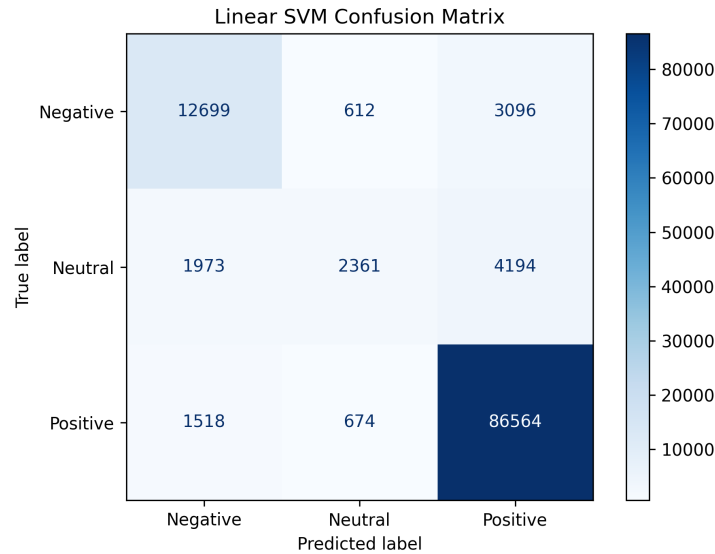


Figure 4: Confusion Matrix of the Linear SVM Model

The confusion matrix indicates that the Linear SVM model correctly classified the majority of customer reviews with relatively few misclassifications, confirming its suitability for sentiment analysis.

8 Conclusion

This project presented an end-to-end sentiment analysis system using Machine Learning and Natural Language Processing techniques. Customer reviews from the Amazon Product Reviews dataset were preprocessed and transformed using TF-IDF Vectorization before training three machine learning models.

Experimental evaluation showed that the Linear SVM model achieved the best classification performance with an accuracy of **89.39%**. The developed system demonstrates how machine learning can efficiently analyze large volumes of customer feedback and support data-driven decision making.

9 Future Scope

The proposed system can be further enhanced by:

- Deploying the trained model as a web application.
- Using Deep Learning models such as LSTM or Transformer-based architectures.
- Extending the system to multilingual sentiment analysis.
- Performing aspect-based sentiment analysis for more detailed customer insights.

10 References

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2. Scikit-learn Developers. *Scikit-learn: Machine Learning in Python*.
3. Steven Bird, Edward Loper, and Ewan Klein. *Natural Language Toolkit (NLTK)*.
4. Amazon Product Reviews Dataset.